COURSE TITLE	**INFORMATION AND WEB SECURITY**
COURSE CODE	**01CT1515**
COURSE CREDITS	**4**

Objective:

1 To provide knowledge regarding working of information and web security mechanisms with by considering security issues and to learn the insights of information assurance techniques in area of information and web security.

Course Outcomes: After completion of this course, student will be able to:

1 Analyze common threats, attack and mechanism, deal with them (Analyze).

2 Apply various Symmetric and asymmetric key Algorithms (Apply).

3 Apply the concepts of Hash function, digital signature and digital certificates (Apply).

4 Understand security concepts for web development (Understand).

5 Identify the various security hazards related to web applications and need of security measures (Analyze).

Pre-requisite of course: Internet & Web Technology

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Introduction To Information Security** Overview of internet security, Firewalls, , Internet security, Management concepts and information privacy and copyright issues, Basics of asymmetric cryptosystems	2
2	**Conventional Encryption & Number Theory:** Conventional encryption model, Steganography, Classical encryption techniques (Substitution and Transposition techniques), , Prime and relative prime numbers, Modular arithmetic, Euler's theorem, , Euclid's algorithm, Discrete logarithmic, Modular arithmetic, The Euclidean algorithm, Finite field of the form	6
3	**Block Cipher Method** Stream ciphers and block ciphers, Block cipher structure, Data encryption standard (DES) with example Strength of DES, Blowfish, RC5, Cast-128, RC2, AES with structure, Traffic confidentiality, Random number generation, Key distribution	6
4	**Advanced symmetric cipher** Multiple encryption and triple DES, Electronic code book, Cipher block chaining mode, Cipher feedback mode	4

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Contents : Unit	Topics	Contact Hours
5	**Message authentication and hash functions:** Authentication requirement, Message authentication code, Hash functions, Security of hash functions and macs, MD5 message digest algorithm, Secure hash algorithm	4
6	**Introduction to Web Security** IP security overview, Architecture, Authentication header, Key management, pretty good privacy, S/Mime and types, Web security requirement, SSL and transport layer security, Secure electronic transactions, Web security requirement, SSL and transport layer security, Secure electronic transactions, Web Threat Model, Authenticated Sessions, Cross-site Scripting, JavaScript Hijacking, Web Service Security, Website Vulnerability Scanner.	4
7	**Web Application Security &Vulnerability** Web Application Security, How Does Web Application Security Work, Web Application Lifecycle Maintenance, Importance of Web Application Security, Broken Access Control, Broken Authentication and Session Management, Buffer Overflows, Cross Site Scripting Flaws, Denial of Service, Improper Error Handling, Insecure Configuration Management, Insecure Storage, SQL Injection Flaws, Unvalidated Input, Defensive Measures	8
8	**E-Commerce Security** Overview of online Security issues, Security for client computers, Communication channel security, Security for server computers	4
9	**Public Key Cryptography** Public key cryptosystems with applications,, Requirements and cryptanalysis, RSA algorithm, Diffie-hillman key exchange algorithm, Man-in-Middle attack	4
Total Hours		**42**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Experiment 1** Write code to implement Substitution cipher and Transposition cipher	2
2	**Experiment 2** Write code to implement DES	2
3	**Experiment 3** Write code to implement AES	2
4	**Experiment 4** Write code to implement Diffie-hillman key exchange algorithm	2
5	**Experiment 5** Write code to implement RSA algorithm	2
6	**Experiment 6** Write code to implement MD5 message digest algorithm	2

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Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
7	**Experiment 7** Write code to implement Dictionary Attack and Brute Force Attack	2
8	**Experiment 8** Installation of Wire shark, tcpdump, etc and observe data transferred in client server communication using UDP/TCP and identify the UDP/TCP datagram	2
9	**Experiment 9** Installation of rootkits and study about the variety of options	2
10	**Experiment 10** Perform an Experiment to Sniff Traffic using ARP Poisoning	2
11	**Experiment 11** Demonstrate intrusion detection system using any tool (snort or any other s/w).	2
12	**Experiment 12** Demonstrate how to provide secure data storage, secure data transmission and for creating digital signatures	2
13	**Experiment 13** Installation and overview of Burpsuite Tools in Kali Linux.	2
14	**Experiment 14** Perform SQL injection	2
15	**Experiment 15** Perform Cross-site scripting and Cross-site request forgery (CSRF)	2
16	**Experiment 16** Demonstrate Information disclosure vulnerabilities	2
17	**Experiment 17** Configure NGINX and Apache webserver in Linux with https protocol	2
18	**Experiment 18** Configure Fail2ban with NGINX to protect the web server.	2
Total Hours		**36**

Textbook :

1 Cryptography & Network Security, Mukhopadhyay, McGraw-Hill, 2005

2 Cryptography and Network Security, Atul Kahate, McGraw-Hill, 2003

References:

1 Cryptography and Network Principles and Practice, Cryptography and Network Principles and Practice, William Stallings, Pearson Publication, 2003

2 Cyber Law and Information Security, Cyber Law and Information Security, Faiyaz Ahamad, Dreamtech Publications, 2009

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
5.00	25.00	30.00	25.00	10.00	5.00

Instructional Method:

1 The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.

2 Practical examination will be conducted at the end of the semester for evaluation of performance of students in laboratory.

3 Students may use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory, etc.

4 The course delivery method will depend upon the requirement of content and need of the students. The teacher in addition to conventional teaching method (Chalk and Talk) may use any of the tools such as demonstration, role play, Quiz, brainstorming, Flipped class, Project based learning, Collaborative learning, MOOCs etc. for effective teaching.

Supplementary Resources:

1 https://portswigger.net/web-security/all-labs#sql-injection

2 https://portswigger.net/web-security/all-labs#cross-site-scripting

3 Students may use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory, etc.

4 The course delivery method will depend upon the requirement of content and need of the students. The teacher in addition to conventional teaching method (Chalk and Talk) may use any of the tools such as demonstration, role play, Quiz, brainstorming, Flipped class, Project based learning, Collaborative learning, MOOCs etc. for effective teaching.

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